How to Make Peace Without Killing Each Other

Sylvain Saucier

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9. Practical Implications: Personal Practice and Collective Institutions

The Cube is designed for both personal and collective use.

Personally, it is a daily practice. Before a difficult conversation, one can ask: Am I keeping my feedback short? Am I curious about the other's experience? Am I protecting the fragile thing in them that might be hurt? The Cube does not guarantee right action, but it makes wrong action harder to sustain, because drift becomes noticeable.

Collectively, institutions can use the Cube as a diagnostic. A legal system can ask whether its procedures shorten or lengthen feedback (e.g., statutes of limitations, accessibility of appeal). An educational curriculum can ask whether it fosters curiosity or mere compliance. An economic policy can ask whether it protects the fragile or only the powerful. In each case, the Cube provides a shared evaluative language that transcends ideological boundaries.

The practice of the Cube is not a one-time conversion. It is a skill, like navigation by the stars. It must be learned, practiced, and recalibrated regularly. The exemplar figures — Jesus, Mohamed, Buddha — serve as "known signals" that verify the compass is still pointing true. If one's practice drifts into cruelty, it has left their gravitational field.

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10. Conclusion – A Non-Ideology for a Pluralistic World

Humanity does not need a single truth. It needs a shared direction.

The Cube offers that direction without asking anyone to abandon their home. It is a meta-tool, a compass that fits in the pocket of every pilgrim, regardless of the path they walk. It defines GOOD not as a doctrine but as a growing diversity of constructive complexity. It points toward GOD not as a jealous deity but as the ultimate emergent property of a universe that has been creatively resolving itself since the first moment.

I invite spiritual leaders, philosophers, scientists, and all people of goodwill to test this compass. Debate it. Refine it. But most of all, use it. Let us shift the dialogue from who is most righteous to what builds the most life. Let us protect the fragile, stay curious, and learn from our mistakes — together.

The constructive attractor does not demand uniformity. It demands only that we all face the same way, each with our own stories, and walk.

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How to Make Peace Without Killing Each Other

Sylvain Saucier

Part I – Why We Keep Killing Each Other

(The Wrong Map)

1. The Mistake of Mirrors

When people talk about making peace, they almost always mean one thing: **balance the forces**.
Give both sides enough weapons that attacking is suicide.
Build opposing armies, equal in strength, frozen in a stalemate.

Call it deterrence, call it power symmetry, call it the balance of terror.
It has a seductive, mechanical logic. If neither side can win, neither side will fight.

And it works—for a while.
A sheet of ice can hold a boulder for a long time, too. But ice isn't structure; it's a temporary phase.

The mistake is to confuse a paused fight with a healed relationship. If I hold a gun to your head and you hold a gun to mine, we have perfect balance. Are we at peace?

The trouble is that this model treats nations, tribes, and peoples as if they were billiard balls: inert objects moved only by external force. It imagines that violence is a rational calculation that can be corrected by adjusting the costs. But humans—and the societies they form—are not billiard balls. They don't just collide; they interpret, they resent, they dream. A balance of power that ignores these inner currents is a dam built over a volcano.

Worse, the mirror-symmetry logic trains everyone to see the world as a zero-sum board. Every gain for them is a loss for us. Every act of kindness is a trick. Every compromise is a prelude to betrayal. Under this logic, even the peacemakers become paranoid.

And the paranoia isn't crazy, because the symmetry game guarantees that the moment one side gains an edge, the other must strike first or perish. So both sides spend their peace preparing for war, which slowly warps their economies, their education, their myths, until war is not an option but an identity.

So the first thing we must do in a book about making peace is to admit that **the strategic balance model is not a path to peace; it is a long-duration fuse**.

This doesn't mean we ignore military realities. It means we stop pretending they are the foundation of peace. The foundation of peace is something else entirely—something that has to do with how living systems regulate themselves. And to understand that, we need to look at what we actually are.

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2. Autoimmune Humanity

We are not a collection of separate billiard balls.
Humanity is a single, sprawling living system.

I mean that literally, not poetically. A city breathes in resources and exhales waste. A nation has a metabolism, a memory, an immune response. The entire human species has become a networked superorganism whose nervous system is the internet, whose circulatory system is global trade, whose gut is industrial agriculture.

And if humanity is a living body, then war is an autoimmune disease.

Consider what happens in an autoimmune disorder. Some part of the body's own tissue gets mistaken for a foreign invader. Perhaps a few cells display an unusual protein, or a signal gets misinterpreted. The immune system, which exists to protect the whole, launches a ferocious attack against itself. It sends killer cells to destroy perfectly healthy tissue. The body fights a war inside its own borders, and often the weapons—inflammation, fever, scarring—do more damage than the initial trigger.

Now look at a war. Two groups of the same species, often genetically indistinguishable, sharing the same planet, the same biology, the same fundamental needs, suddenly decide the other is a parasite that must be eliminated. The "impurity" is usually a story—a difference of flag, of book, of ritual, of historical grievance. The immune system of the group tags the neighbor as "non-self" and mobilizes to purge it. The weapons—bombs, sanctions, hate propaganda—destroy the shared infrastructure, the shared ecology, the shared future.

The Cube replaces this with a shared inquiry: **What are the systemic consequences of this teaching or practice?**

Spiritual leaders from different traditions can sit together and ask: - Does this doctrine, if widely adopted, shorten or lengthen our ability to hear and correct harm? - Does it open or close the exploration of life's deepest questions? - Does it shield or endanger the fragile among us and in creation?

Suddenly, a Buddhist monk, a Muslim imam, and a Christian theologian are not comparing scriptures; they are comparing *systemic outcomes*. They can disagree profoundly on metaphysics and still converge on a shared diagnosis: a practice that silences victims, forbids questions, or tramples minorities is misaligned with the constructive attractor, no matter which holy book it claims to quote.

This is not syncretism. It does not ask anyone to dilute their faith. It asks only that each tradition use the same compass to self-regulate, and that they hold each other accountable using a language of consequence rather than righteousness. The goal is not one world religion, but a world where many religions all face the same attractor — each preserving its unique genius, each contributing to a global ecology of constructive complexity.

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8. Philosophical Grounding: Cybernetics, Process, and Self-Referential Topology

The Cube is not an arbitrary invention. It is a direct application of the logic explored in *Time as Dice*, where I argued that any sufficiently self-referential topology is logically underdetermined — its future states are created in time, not deduced from a static description.

An ethical system for such a universe cannot be a set of deterministic rules. It must be a **soft** framework — one that provides orienting constraints without prescribing outcomes. The Cube is precisely that: a constraint network of three mutual dependencies. Feedback without curiosity becomes stale; curiosity without respect becomes predatory; respect without feedback becomes blind sentimentality. They regulate each other, and the user's practice is the real-time resolution of those constraints into a lived path.

In cybernetic terms, the Cube is a homeostat for the moral imagination. In process philosophy terms, it is a grammar of actual occasions — a way to prehend the world so that each concrete decision contributes to the growing intensity of experience that Whitehead called "the good." And in spiritual terms, it is the geometry of sanctity.

- `(+--)` Feedback without curiosity or respect: a rigid surveillance state.

These are not damnable states; they are *trade-off zones*. Every person and institution will spend time in them. The Cube merely reveals the direction of correction. It does not judge; it orients.

The Compass, Not a Map

Crucially, the Cube does not tell a user *what to do* in a concrete situation. It provides no list of permitted and forbidden acts. It is a compass, not a map. Learning to use it regularly — consulting it in moments of decision, reflecting on past actions — builds the skill of ethical navigation. Over time, the user internalizes the pull of the `(+++)` attractor and becomes more sensitive to drift.

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6. Constructive Complexity and the Constructor Attractor

The term "constructive" in the definition of GOOD is deliberately recursive. It means: tending toward the `(+++)` attractor. This self-reference is not a flaw; it is the signature of a living, self-regulating system. Just as the `Seedy64` device's output cannot be predicted from its initial zero state, the specific path toward GOOD cannot be prescribed in advance. It must be generated in real time by the user's own practice.

Historical exemplars — Jesus, Mohamed, Buddha — are not figures who left static rulebooks. They lived in such deep alignment with the constructive attractor that their lives became reference points for the whole of humanity. They embodied feedback (they listened to outcasts, challenged hypocrisy), curiosity (they upended tradition, sought solitude and revelation), and respect (they protected the poor, the sick, the sinner). To follow them is not to mimic their specific acts, but to align with the same attractor they inhabited so fully.

The Cube does not replace these figures; it helps anyone — from any tradition — see *why* they are attractors, and how to orient toward the same underlying dynamics.

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7. Application to Interfaith Dialogue and Systemic Consequences

The Cube's most urgent application is to the relationship between religions and cultures. Currently, much interfaith dialogue is a contest of righteousness: "my revelation is complete, yours is partial." This is a zero-sum game that feeds the destructive attractor. It shortens feedback (by dismissing outside critique), suppresses curiosity (by protecting dogma), and endangers the fragile (by justifying domination).

If you could zoom out and see the whole Earth as one organism, every bombing run would look like a hand attacking a foot. Every genocide would look like a spleen trying to destroy the liver because the liver sang the wrong song.

The autoimmune analogy is not just a clever image. It suggests a whole different way of thinking about what peace requires. You don't cure lupus by putting half the immune system on one side of a room and the other half on the other side and telling them to stare at each other until they calm down. You don't treat rheumatoid arthritis by giving each inflamed joint its own arsenal so that no joint can overpower the others. That would be absurd.

Yet that's exactly what the balance-of-power doctrine prescribes: an armed truce between two inflamed identities.

Instead, to heal an autoimmune condition, you must go to the root: the failure of the immune system to recognize "self." You have to rebuild the signaling pathways that say *this is us, this is safe, this belongs*. You calm the overactive attack, not by threatening it with counter-attack, but by restoring the regulatory mechanisms that keep the system whole.

Applied to humanity, that means the real work of peace is restoring the capacity of one part of the superorganism to recognize another part as itself. It means rebuilding the feedback loops that would normally tell a nation, a tribe, a religion that the people across the river are not a virus—they are a limb, and if you cut them off, you bleed.

So if we want to make peace without killing each other, we must stop thinking of peace as an external ceasefire and start thinking of it as an internal healing.

That shift in perspective brings us to the most stubborn obstacle of all: the way we think about the future itself.

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3. The Demon's Conceit

There is an old fantasy in science and philosophy.
Laplace, an 18th-century mathematician, imagined a super-intelligence that knew the position and velocity of every particle in the universe, and all the laws of physics. For such an intellect, nothing would be uncertain. The whole future and the whole past would be present before its eyes.

Laplace's demon.

The demon is the ultimate expression of determinism—the belief that the future is already fixed, like a movie already filmed, and we are merely watching it play out frame by frame.

This belief is not just an abstract philosophical position. It has leaked into our politics, our psychology, our peacemaking. When a conflict seems eternal—say, Israelis and Palestinians, or Hindus and Muslims, or the Cold War—people start to talk as if the conflict is "inevitable." They speak of cycles of violence, of deep-seated hatreds, of the impossibility of resolution. They become deterministic: the future is already written, and it's written in blood.

And that belief is the greatest enemy of peace. Because if the future is fixed, why try? If the next war is inevitable, why not just prepare for it now and at least be the winner? Determinism of the mind breeds passivity of the will.

But Laplace's demon has been quietly refuted, not by philosophers, but by a tiny computer program I wrote called the Seedy device.

The Seedy Argument in Plain Words

Imagine three ants running in a circle on your desk. Each ant watches the ant ahead of it. When it sees the other ant move, it does a little calculation and then takes a step, which the next ant then watches. You start all three ants at exactly the same spot, with exactly the same instructions, and you let them go.

What happens? You'd think, because they start identically, they'd all march in perfect lockstep forever. But they don't. They quickly fall into a chaotic, unpredictable dance. In fact, if you run the experiment again from the exact same starting point, you get a completely different dance. And if you only watch the very first step of each run, you'll eventually see *every possible first step* that the three ant's rules allow.

How is that possible? The instructions are deterministic. The starting point is identical. There's no coin flip. The magic is that the three ants form a loop: A depends on B, B depends on C, C depends on A, all at the same instant. That loop cannot be untangled on paper. To know what A will do, you must first know what B will do, which requires knowing what C will do, which requires knowing what A will do. The equations cannot be solved in advance. The only way to find out what happens is to actually run the system—to let time and physics resolve the mutual dependencies, consuming real energy and producing a unique outcome that simply did not exist before.

The Seedy device shows that even in a perfectly deterministic world, a system with self-reference has no predetermined future. The future is not a train track; it is a cloud of possibilities, a cone of probability. Most of the time, without any special push, the system settles into the fat center of that cone—the boring, "most probable" path. But the edges of the cone, the unlikely outcomes, are just as real.

Why does this matter for peace? Because our current "inevitable vector" toward further conflict is nothing more than the fat center of the cone. It's the easiest resolution of all our current social loops—but it's not the only possible resolution.

Axis 3: Respect and the Protection of the Fragile

Respect is the protection of the fragile.

In any complex system, certain configurations are rare, delicate, and easily destroyed. They carry the seeds of future diversity. A child's trust, a minority language, a nascent idea, a vulnerable ecosystem, a dissenting voice — these are the fragile carriers of novelty. A system that optimizes itself for efficiency or purity will crush them. Respect is the deliberate, counter-optimization force that shields the fragile from the steamroller of the majority.

This axis is the deepest root of what traditions call compassion, mercy, or ahimsa. It is the recognition that the good is not always the strongest, and that the health of the whole depends on the protection of the weak.

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5. The Cube as a Navigational Tool

The three axes define a three-dimensional space. Any action, policy, doctrine, or personal choice can be roughly located in this space by assessing whether it increases or decreases feedback, curiosity, and respect.

The Constructive Attractor (+++)

The corner (+Feedback, +Curiosity, +Respect) is the **constructive attractor**. To move toward this corner is to increase constructive complexity. It is the direction of GOOD.

The Destructive Attractor (---)

The opposite corner (-Feedback, -Curiosity, -Respect) is the **destructive attractor**. It represents a system that has shut down all self-correction, killed all exploration, and obliterated the fragile. It is the direction of collapse, tyranny, and death.

The Six Fuzzy Middles

The six remaining combinations are the real terrain of ethical life:

- (--+) No feedback, no curiosity, but some respect: a stagnant but gentle tradition.
- (-++) No feedback, but curious and respectful: an aimless openness.
- (+-+) Feedback and respect without curiosity: a well-run but closed orthodoxy.
- (++-) Feedback and curiosity without respect: a ruthless, experimental technocracy.
- (-+-) No feedback, some curiosity, no respect: a chaotic exploitation.

resolutions — the forever-growing complexity of what has become — is GOD. In Whiteheadian terms, this is the consequent nature of God. In cybernetic terms, it is the global attractor of the cosmic process. I leave it to each tradition to name this horizon in its own language.

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4. The Three Axes – Cybernetic Foundations

The Cube is built on three orthogonal axes. Each axis is not a moral commandment but a *cybernetic condition* that any viable, evolving system must satisfy. They are derived from the minimal requirements for a self-referential topology to remain creative rather than collapsing into a destructive attractor.

Axis 1: Feedback and Learning

> **Complex systems must maintain short feedback loops and learn from the consequences of their actions.**

In cybernetics, feedback is the mechanism by which a system corrects its trajectory. Short feedback loops mean that consequences are observed quickly and the system adjusts in near-real time. When feedback loops are long, corrupted, or severed, the system becomes blind. It accumulates error, drifts into pathology, and eventually destroys itself.

For a religion or culture, this means: institutions must listen to those they affect, measure outcomes honestly, and correct course when harm is observed. No doctrine is immune to revision if its application produces suffering. This is not a rejection of tradition; it is the mechanism by which a living tradition survives.

Axis 2: Curiosity and Exploration

> **Curiosity is the exploration of solution space.**

A system that does not explore its possible states is brittle. It cannot adapt to changing environments. Curiosity is the drive to probe the unknown, to ask uncomfortable questions, to entertain heresy in the protected space of thought before judging it in practice. Without curiosity, a culture becomes a monoculture, and a monoculture is an extinction risk.

Spiritually, curiosity is the open hand toward mystery. It is the recognition that the divine, however conceived, is always larger than our current understanding. Every great exemplar — Jesus questioning the Pharisees, Mohamed receiving revelation in a cave, Buddha leaving the palace — began with an act of radical curiosity.

And here's the pragmatic twist: we, as conscious parts of the system, can deliberately skew the resolution. We can exert effort, burn energy, change the loops so that an outcome that was merely a thin sliver on the edge becomes the new center.

Determinism is false. The future is not yet written. And that means peace is always possible, no matter how stacked the deck appears.

So far, we've identified three fatal errors that keep us killing each other:
1. We think peace is a military balance, when it's not.
2. We treat ourselves as separate objects, when we are a single organism with an autoimmune condition.
3. We believe the future is fixed, when it's an open cone we can tilt.

Now let's build the correct map—the map that treats humanity as a living, creative, self-healing system and shows us how to navigate toward health.

Part II – The Real Anatomy of Peace

(A Living Map)

4. The Social Organism

Everywhere you look, life organizes into larger lives.
Single cells joined into multicellular bodies. Individual bees joined into a hive mind. Individual humans joined into families, tribes, cities, nations, civilizations.

These larger entities are not just metaphors. They satisfy every scientific definition of a living system:

- They maintain their internal order despite external chaos (homeostasis).
- They take in energy and matter and excrete waste (metabolism).
- They respond to their environment and adapt (responsiveness).
- They grow, change, and occasionally reproduce (development and reproduction).
- They are composed of smaller living units that sacrifice some autonomy for the good of the whole (hierarchical organization).

A forest is a slow, green animal. A city is a buzzing hive of concrete and flesh. A religion is a social organism that spans centuries and continents, with a genome of stories and rituals.

When you start seeing humanity this way, some stubborn old questions dissolve.

"Why can't we all just get along?" Because a living body isn't a democratic assembly; it's a balancing act of thousands of competing subsystems—hormones, nerves, organs, microbes—each with its own local interests. The heart wants blood pressure high enough to pump; the kidneys want it low enough to filter. They don't "agree"; they regulate each other through feedback.

"Is conflict natural?" Yes, as natural as the competition between the immune system and a pathogen, or between the need for sleep and the need to hunt. The question isn't whether conflict exists; it's whether the larger body has the regulatory mechanisms to keep conflict constructive rather than destructive.

"What is a nation, really?" A nation is a shared story that coordinates millions of individual metabolisms into a coherent collective action. When the story is healthy, it enables cooperation at scale. When the story becomes rigid and closed—an ideology that refuses feedback—it becomes a tumor.

So the first principle of the living map is: **The unit of peace is not the individual, not the state, but the health of the whole superorganism.**

And the second principle follows directly: **To heal the whole, we must understand the vital signs of a living society.**

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5. Three Vitals of Any Living Society

Every living system, from a single cell to a global civilization, depends on three deep competencies. Think of them as the vital signs a doctor checks: heart rate, blood pressure, temperature. If any one of them crashes, the organism dies.

For a social organism, the three vitals are:

1. Feedback and Learning
The ability to feel the consequences of your actions and adjust.

A body without a nervous system cannot sense injury; it will blunder into a fire and burn without flinching. A society without feedback cannot see the harm its policies cause. It becomes deaf and blind.

Feedback means: when a law creates suffering, that suffering must be visible to the lawmakers, quickly and clearly. When a company pollutes a river, the people downstream must have a powerful

claims to deduce the right future from a fixed set of principles. This is the ideological form of Laplace's demon: a belief that the good can be specified once and for all, and that deviation is error.

The historical record shows the cost. When an ideology hardens, it shortens its own feedback loops (it stops listening), suppresses curiosity (it punishes questions), and sacrifices the fragile (minorities, dissenters, the vulnerable) for the sake of purity. In the language of my earlier work on self-referential topologies (*Time as Dice*), a closed ideology is a deterministic system trying to function in a world that is irreducibly creative. It cannot succeed, and in failing, it destroys the diversity that makes the world resilient.

What is needed is not a better ideology, but a **meta-compass**: a tool that helps any person, any culture, any religion navigate toward health without forcing them to abandon their identity. This paper proposes such a tool — the Cube — built from three cybernetic principles, and argues that it can serve as the common non-ideology for a pluralistic world.

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3. Definitions: GOOD and GOD

All ethical systems require an orienting concept of the good. I propose a definition that is inherently dynamic and systemic:

> **GOOD** is a growing diversity of constructive complexity.

Three terms matter here:

- **Diversity**: heterogeneity, the presence of many distinct forms of life, thought, and culture.
- **Complexity**: the richness of interconnections and mutual dependencies within and between systems.
- **Constructive**: that which moves a system toward the (+++) attractor of the Cube — i.e., complexity that maintains short feedback loops, fosters curiosity, and protects the fragile. The word is thus both intuitive and technically anchored.

GOOD, so defined, is not a static endpoint. It is a *direction*. A system is good not because it has reached perfection, but because it is tending toward richer, more connected, more self-regulating diversity.

If GOOD is the direction, then **GOD** is the ultimate horizon:

> **GOD** is the ultimate emergent property of the universe.

This is not a theological claim about a personal deity — though it is entirely compatible with such claims. It is, rather, a structural statement: if the universe is a self-referential topology that has been resolving its own constraints since the Big Bang, then the accumulated totality of all those creative

1. Introduction

Imagine you are walking in a vast, dark forest. You have no map, and the trees all look the same. Somewhere, you know, there is a clearing where life is safe and good. But the forest is full of voices, and every voice claims to know the one true path. One voice says, "Go north, always north, and never question why." Another says, "Go south, and if you meet someone going north, they are your enemy." A third says, "Stay exactly where you are; movement is sin." All of them sound certain, and all of them contradict each other.

You do not need a louder voice. You need a compass.

A compass is not a map. It does not tell you the exact path. It does not know the forest. It only points to magnetic north—a direction that is true regardless of where you stand or what you believe. If you have a compass, you can still get lost, you can still stumble, but you can always re□orient. You can even walk side by side with someone who holds a different map, as long as you both agree which way north is.

Humanity is in just such a forest today. We are divided not only by borders and resources but by ideologies, each claiming to be the complete map. Religions, political systems, cultural identities—each began as a creative way to navigate a particular time and place, but many have hardened into closed systems that can no longer correct themselves. They stop listening. They punish curiosity. They trample the fragile. In the language of cybernetics, they have lost their feedback loops, their capacity to learn, and their care for the delicate carriers of future life.

This paper does not offer a new map. It offers a compass—a simple, three□dimensional ethical navigation tool built from the minimal requirements that any healthy, living system must satisfy. I call it the Cube. The Cube does not ask anyone to abandon their religion, their culture, or their deepest convictions. It asks only that they orient themselves, day by day, toward what I call GOOD: a growing diversity of constructive complexity. And it points, always, toward a horizon that every tradition has glimpsed in its own language—what I name GOD: the ultimate emergent property of a universe that has been creatively resolving itself since the very beginning.

The rest of this paper explains the Cube's three axes, its constructive attractor, and how spiritual leaders and ordinary people alike can use it to move beyond contests of righteousness and instead debate the systemic consequences of their beliefs. It is, in essence, an invitation to stop comparing maps and start sharing a direction.

2. The Problem of Ideology

Every ideology, religious or secular, begins as a creative resolution to a specific set of human constraints. Over time, however, it tends to ossify. It becomes a closed system of prescriptions that

voice that reaches the boardroom. When a dominant cultural group marginalizes a minority, the minority's pain must be felt as a wound in the whole social body.

Short feedback loops are the immune system of society. Long, broken, or corrupted feedback loops are the start of every cancer.

2. Curiosity and Exploration
The drive to understand what you do not yet understand.

A body that cannot explore its environment cannot find food. A mind that cannot question its own assumptions cannot learn. A society that silences dissent, bans books, or declares certain subjects off-limits is a society amputating its own sensory organs.

Curiosity is not a luxury for the leisurely. It is the mechanism by which a system discovers new possibilities, adapts to changing circumstances, and avoids the brittle monoculture that a single shock can shatter.

Spiritually, curiosity is the open hand toward mystery. Every great breakthrough—moral, scientific, artistic—began with someone daring to ask a question that the orthodoxy of the day considered dangerous.

3. Respect and the Protection of the Fragile
The deliberate care for what is easily broken.

In any complex system, certain things are precious and rare: a child's trust, a minority language, a delicate ecosystem, a dissident voice, a pregnant possibility. They are easily crushed by the efficient machinery of the majority. They carry the seeds of future diversity.

Respect is the force that pushes back against optimization. It insists that some things must be protected even when they are not powerful, not profitable, not popular. It is the root of compassion, mercy, ahimsa, and the golden rule.

A society that loses respect becomes a brutal machine that grinds its own weakest parts into dust—and soon discovers that the dust was load-bearing.

These three vitals—feedback, curiosity, respect—are not moral suggestions. They are the non-negotiable requirements for any long-lived, healthy, evolving system.
When they all function strongly, the social organism hums with life. When one or more are broken, the organism slides toward disease.

And the direction of disease—or health—can be mapped.

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6. The Color of Health

Imagine a three-axis space. One axis measures feedback (from short and honest, to long and corrupted). Another measures curiosity (from open and playful, to closed and dogmatic). The third measures respect (from fierce protection of the fragile, to ruthless sacrifice of the weak).

Any society, institution, movement, or even individual life can be located somewhere in this cube.

There is one corner where all three vitals are maximally healthy:
(+Feedback, +Curiosity, +Respect). Call this the **Constructive Attractor**.

There is the opposite corner where all three are maximally broken:
(-Feedback, -Curiosity, -Respect). Call this the **Destructive Attractor**.

The Constructive Attractor is the direction of GOOD. It is where complexity grows, diversity flourishes, and the future becomes richer.
The Destructive Attractor is the direction of collapse, tyranny, and death. It is where life simplifies into ash.

Between these two poles are the fuzzy middles—the various trade-off zones where real human life actually happens. A rigid but gentle tradition might have low curiosity but high respect. A ruthless technocracy might have high feedback and curiosity but no protection for the fragile. None of these are hell; they are simply states that signal a need for correction.

The crucial insight is that these attractors are not just descriptions; they are *forces*.
A system drifting toward the Destructive Attractor will actively resist feedback, punish curiosity, and sacrifice the fragile. It will create ideologies that justify its drift.
A system moving toward the Constructive Attractor will seek out feedback, encourage questions, and build shelters for the vulnerable.

The question for peacemaking then becomes not “Who is right?” but “In which direction is this action, this policy, this story pulling the whole?”

This is a practical tool. When a leader proposes a wall, ask: Does this shorten feedback between the two sides? Does it open or close curiosity about the other? Does it shield the fragile or sacrifice them for a sense of security?

When a peace agreement is on the table, ask the same questions. Don’t just ask if it distributes land or weapons fairly. Ask if it builds mechanisms that will shorten feedback loops between the former enemies, foster curiosity about each other’s experience, and protect the fragile on both sides from the backwash of the peace.

The map is simple. The navigation is the work of a lifetime.

A Cybernetic Compass for Ethical Navigation and Interfaith Unity

Author: Sylvain Saucier
Repository: github.com/sysaulab/softrng
Precursor work: *Time as Dice: A Computational Refutation of Laplacean Determinism*

Abstract

Humanity is fractured not only by conflicting interests but by competing ideologies, each claiming final knowledge of the good. Yet every closed ideology, when it hardens, becomes a deterministic machine that crushes the very diversity it claims to serve. I propose a meta-ideology — a non-ideology — grounded in cybernetics and the logic of self-referential systems. At its core is a simple three-axis cube: **feedback and learning**, **curiosity and exploration**, and **respect for the fragile**. These axes define a navigational space with a constructive attractor `(+++)` and a destructive attractor `(---)`. I define **GOOD** as a growing diversity of constructive complexity, and **GOD** as the ultimate emergent property of the universe — the forever-receding limit toward which constructive complexity tends. The cube serves as a compass, not a map. It does not prescribe what to believe, but how to orient any belief system toward systemic health. I argue that spiritual leaders from all traditions can use this compass to move beyond contests of righteousness and instead debate the *systemic consequences* of their theologies. In doing so, they can unite humanity under different religions and cultures, each preserving its unique identity while contributing to a shared constructive attractor. This paper provides the philosophical grounding, the cybernetic justification, and a practical invitation.

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7. Time as Dice

We already met the Seedy device, but now we must place it at the heart of the living map.

The living-systems view tells us that the future is not predetermined; it is a probability cone. The fat center of the cone is what will happen if we change nothing—the path of least resistance, the default resolution of all our current mutual dependencies.

But the edges of the cone are not miracles. They are genuine potential futures, just less probable under current conditions.

Think of a war-prone region as a ring of three ants: multiple groups with mutual dependencies, mutual narratives, mutual provocations. The default resolution of their loops—the fat center—may be another cycle of retaliation. But that resolution is not fated. It is simply the easiest one for the loops to fall into, given their current configuration.

If you change the configuration—if you add a new feedback loop, introduce a shock of curiosity, protect a fragile bridge-building initiative—you are, in effect, rewriting the rules the ants are running. You tilt the cone. An outcome that was a 1% probability can become a 30% probability, then an 80% probability.

This is not wishful thinking. It's a direct consequence of how self-referential systems resolve. The Seedy device is proof that from the exact same starting conditions, completely different outcomes are possible. And it is also proof that the system can be nudged toward those outcomes by deliberate intervention.

That's what free will is, in a cybernetic sense: not a magical escape from causality, but the capacity of a self-referent system to participate in its own resolution. We are part of the loop. Our choices become new constraints that steer the resolution.

Therefore, despair is not realism; it's a mathematical mistake. It confuses the fat center with the whole cone. And it is that confusion—that false determinism—that often prevents us from making the very interventions that could tilt the cone toward peace.

So now, with the map in hand, we can turn to the practical art of tilting.

Part III – Freeing the Will, Tending the Body

(How to Skew the Cone)

8. The Edge of the Cone

Before we get into the mechanics, we must take a moment for hope. Not sentimentality—strategic hope.

History is littered with moments where the fat center of the cone seemed to make disaster inevitable, and yet the disaster was averted by a thin-slice intervention.

- The Cuban Missile Crisis: two superpowers locked in a deterministic spiral toward nuclear exchange. The resolution came not from military symmetry (the balance was already there) but from backchannel feedback, curiosity about the other's actual fear, and a decision to protect the fragile—the children, the cities, the future.
- South Africa's transition from apartheid: the fat center of the cone was a bloodbath. The actual resolution was tilted by truth-telling (feedback), the willingness to imagine a shared nation (curiosity), and the painstaking protection of fragile bridge-builders on all sides (respect).
- The end of the Cold War: not through victory but through a cascade of feedback—ordinary people refusing to be lied to, a leader (Gorbachev) willing to explore radical curiosity about alternative systems, and, though imperfect, a moment of mutual respect that allowed the dissolution of an empire without a single shot between the superpowers.

In each case, the peacemakers did not have perfect information or guaranteed success. They simply recognized that the default future was not the only future, and they took actions that shifted the odds.

The practical lesson: you do not need to guarantee peace. You only need to shift the probability enough that the peaceful resolution becomes more attractive, more stable, more alive than the old violent loop.

How do you do that? By deliberately strengthening the three vitals in the conflict system.

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9. Shortening the Loop

The single most powerful action you can take toward peace is to **shorten the feedback loop between potential adversaries**.

Most violent conflicts are sustained by information bubbles. Each side lives in a media environment that filters out the other's humanity, that amplifies grievances, that makes the other's pain invisible. The loop is long and corrupted.

To shorten it:

- Create spaces where the stories of the vulnerable are heard directly—not through intermediaries who spin them.
- Support independent journalism, citizen reporting, and human-rights monitoring that track the real

10. Conclusion: The Grammar of Becoming

We began with three friends shouting numbers into a circle, unable to decide what to say until they said it. We end with a cosmos that does the same, from its first fiery moments to the quiet birth of a thought. A generative topology is the formal name for this ancient, ubiquitous pattern: a self□referential loop placed in a physical medium, forced to resolve, and producing genuine novelty as a necessary consequence.

The `Seedy64` device offers an empirical anchor—a minimal, reproducible proof that such systems exist and behave exactly as the theory predicts. From this anchor, we can stretch the concept into biology, where the embryo becomes a generative process, not a blueprint execution; into cognitive science, where the mind becomes a creative resolver, not a clockwork processor; and into cosmology, where the Big Bang becomes the first generative act, not the first domino.

We are not cosmic accidents. We are the inevitable, unpredictable, and genuinely novel outcomes of a topology that had to become. The dice were always there, thrown not by a hidden player but by the very grammar of existence. And that grammar says: *to be is to resolve, and to resolve is to create.*

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8. The Cosmic Generative Topology: The Big Bang as First Resolution

At the other end of the scale, the early universe can be understood as the ultimate generative topology. In the Planck era, all degrees of freedom—spacetime geometry, quantum fields, nascent matter—were in direct causal contact, forming a maximally interconnected, self□referential network. There was no external clock; time itself was part of the system's internal dynamics. The "initial condition," if such a term even applies, was a set of mutual constraints with no unique resolution.

The expansion of the universe provided the "runtime"—the physical process of serialising those constraints into a coherent history. Symmetry breakings, decoherence events, and the emergence of classical structure were the first generative acts. The cosmic microwave background and the large□scale galaxy distribution are the frozen□in record of those first draws. The universe, from its very origin, was a generative topology that had to play dice, not because a random force was inserted, but because a self□referential birth cannot be statically determined.

Thus, quantum indeterminacy itself may be reinterpreted not as a fundamental law but as the low□energy remnant of a primordial generative topology—an ongoing hum of resolution events that never fully settled because the original constraint network was never "solved," only cooled and expanded.

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9. The Nature of Time and the Refutation of Laplace

If generative topologies permeate reality, then time is more than a parameter in equations. Time is the medium of constraint resolution. The arrow of time is not merely thermodynamic; it is *generative*—it marks the irreversible accumulation of novel states that did not exist before they were enacted.

Laplace's demon is refuted not by invoking quantum mechanics or chaotic unpredictability, but by pointing out that the demon's required snapshot is logically incomplete for any system containing a generative topology. The demon would need to know the entire serialization history in advance, but that history is the very thing it is trying to predict. It would have to become the universe, not simulate it. Determinism is not false; it is *inapplicable* to systems whose initial specifications are underdetermined.

The universe plays dice. It plays dice necessarily, because its origin, its living processes, and its conscious beings are generative topologies that must resolve themselves in time. Dice are not an amendment to law; they are the structural signature of a reality that cannot help but create.

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consequences of policies on both sides.
- Design governance systems where decision-makers are forced to face the communities their decisions affect, in person, regularly—not just through polls or reports.
- In your own life, refuse the convenience of not knowing. If a product you buy is made by exploited labor, find out. If a policy you support hurts a group you never meet, go meet them.

Short feedback loops are uncomfortable. They burn energy. That's the point. The discomfort is the body's signal that something is wrong. Numbness is not peace; it's pre-gangrene.

At the international level, this means things like:
- Truth and reconciliation commissions that make perpetrators face victims.
- Cross-border community exchanges that let ordinary people see each other as neighbors, not news headlines.
- Rapid-response conflict early-warning systems that make escalation visible before it's too late.

Every time you shorten a loop, you reduce the chance that the social body will attack itself blindly.

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10. Re-complicating the Image

The second lever is curiosity.

War propaganda always does one thing: it simplifies the enemy into a cartoon. A single, hateful, subhuman face. That simplification is a psychological necessity because it's hard to kill a complex human being with a family, a sense of humor, and a favorite song.

So the peacemaker's counter-move is to re-complicate the image. Not to excuse atrocities, but to restore the full humanity of the other, because only then can real negotiation, real empathy, real peace emerge.

Practically:
- Support art, film, and literature that crosses the lines of conflict and tells the stories that the official narratives suppress.
- In education, teach multiple perspectives on history, including the perspectives of former enemies. Not to relativize truth, but to expand it.
- In dialogue, forbid the use of dehumanizing labels. Insist on naming the specific actions and the specific people, not the caricatures.
- Cultivate the spiritual practice of radical curiosity: the discipline of asking "What does the world look like from their side?" without immediately rushing to judge.

Curiosity is not a betrayal of your own side. It's an intelligence-gathering operation that reveals the true terrain of the conflict. And very often, it discovers that the enemy's fears and hopes are not so

different from your own.

That discovery is the beginning of the end of war.

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11. Shielding the Fragile

The third lever is respect.

In every conflict, there are fragile things that can tip the balance toward health or destruction: the moderate voices, the cross-community families, the children, the ecosystems, the nascent institutions of cooperation.

A society sinking toward the Destructive Attractor will systematically crush these fragile things. It will label peace-seekers as traitors, starve trans-border economies, poison shared water supplies, and sacrifice the young to feed the old hatred.

A peacemaker's task is to identify the fragile and build a shield around it.
- Protect dissidents and bridge-builders from persecution, using international pressure, safe havens, and public solidarity.
- Invest in the ecological commons—shared rivers, shared air, shared climate—because a dying environment can push any tension into war.
- Prioritize the well-being of children on all sides, because children's memories are the seeds of the next generation's either peace or revenge.
- Defend minority cultures and languages as treasures of the whole human organism, not as obstacles to national unity.

Respect is not always passive. Sometimes it requires fierce protection—the kind of fierce that refuses to let the powerful trample the weak. But the fierceness is on behalf of life, not against life.

And when all three levers—feedback, curiosity, respect—are pulled together, the cone tilts. Sometimes slowly, sometimes suddenly. But it tilts.

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12. Immune Therapies

It helps to see these principles in action. History is full of peace movements that, knowingly or not, acted as the social body's own regulatory system, calming the autoimmune storm.

Consider **Gandhi's satyagraha**. It worked not because the British Empire suddenly developed a conscience, but because Gandhi systematically shortened the feedback loop. He forced the Empire to see the consequences of its rule—the salt marches, the boycotts, the spectacle of nonviolent

This explains why identical twins, sharing the same genome and often the same uterine environment, are not truly identical. They are different first draws from the same generative topology. Their phenotypic differences—fingerprints, neural wiring, organ quirks—are not errors in a copying process; they are the natural, expected output of a system that cannot produce the same output twice because its output is not stored anywhere. It is enacted.

The evolutionary implication is profound. Variation is not solely the product of random DNA mutations; it is an ongoing, active output of the developmental generative topology. Natural selection filters the novelty that development constantly creates, much as a statistical test assesses the output of `Seedy64`. Evolution, in this light, is a creative process channelled by selection, not a blind dice thrower.

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7. The Cognitive Generative Topology: Mind as Active Resolution

If an embryo is a generative topology, so is a conscious brain. The neural architecture of the brain forms a massively recurrent network. Perception influences memory, memory biases expectation, expectation directs attention, and attention shapes perception—a closed loop of mutual dependence. Sensory input does not break this closure; it injects perturbations that are immediately folded into the ongoing resolution process. The brain's connectome is the self□referential rule set; the electrochemical, synaptic, and metabolic milieu is the physical runtime.

A thought, then, is a resolution event. You cannot predict your own next thought by examining a snapshot of your brain's current state, because that state is a set of constraints with no unique successor. The act of thinking—the actual physical propagation of spikes, the settling of attractor networks—is the generation of the thought. This is why genuine insight feels like a discovery, not a recall. It is novel information brought into existence by the brain's generative operation.

The same logic applies to decision and free will. A deliberating mind holds desires, beliefs, values, and perceptions in mutual tension. The constraint network is underdetermined; multiple choices are consistent with the present state. The act of deciding is the physical resolution of that network, a creative event that consumes metabolic energy and produces a new state that was not predetermined. Free will, on this view, is not an uncaused miracle nor mere randomness; it is the active creative novelty of a cognitive generative topology.

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code–machine system. The code provides the self-referential structure; the machine provides the temporal resolution channel. Novelty is generated at their interface.

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6. The Biological Generative Topology: Genome as Code, Cytoplasm as Machine

6.1 The Self-Referential Genome

A fertilised egg contains a genome and a complex cytoplasmic environment. The genome encodes not a linear program but a dense web of regulatory interactions: genes produce proteins that bind to DNA and alter the expression of other genes, often including themselves. Transcription factors form feedback loops, cross□antagonisms, and multi□stable switches. This network is, in the language of Section 3, a self□referential rule set. Its dependency graph is rich in cycles.

The DNA sequence plus the initial concentration of maternal factors is the "code." But just as the `Seedy64` code in isolation does not determine an output, the genome in isolation does not determine a phenotype. It contains constraints, but those constraints admit many possible consistent resolutions—many viable body plans, many cell-fate patterns.

6.2 The Cellular Runtime

The cytoplasm provides the physical runtime. Stochastic gene expression, variable diffusion of morphogens, mechanical forces from cell division and migration, metabolic gradients, and thermal noise all contribute to an environment where the execution of the genetic rules is not sequential and ideal but concurrent, asynchronous, and irreproducible. Cell divisions are not perfectly synchronised; signalling molecules arrive at slightly different times; a given transcription factor may bind a promoter or not depending on local molecular crowding. These micro□events are the analogue of the CPU's cache coherency traffic: they are the "interleavings" that resolve the mutual constraints of the regulatory network.

6.3 The Organism as a Generative Output

The developing embryo is therefore a generative topology in precisely the sense defined. The self□referential gene network (code) coupled to the noisy, physical, temporally extended cellular environment (machine) yields a system whose future states are underdetermined by its initial description. The only way to "know" the phenotype is to let the embryo grow. The body is not computed; it is generated.

suffering—making the cost of oppression impossible to ignore. He re-complexified the image of Indians from "subject race" to morally powerful agents. And he fiercely protected the fragile—insisting that the movement remain nonviolent, defending the dignity of the weakest, refusing to sacrifice the vulnerable for tactical gain.

Or take the **Northern Ireland peace process**. Decades of tit-for-tat violence seemed deterministic. The turning point was a slow, painful knitting of feedback loops (the Good Friday Agreement's cross-community institutions), a re-complicating of narratives (the admission that both sides had grievances and both sides had blood on their hands), and a deliberate protection of the fragile ceasefire moments against spoilers.

The **anti-apartheid movement** succeeded not just by isolating the regime economically, but by building a global feedback system that made white South Africans feel the world's repulsion, by fostering curiosity through cultural boycotts that broke the cultural bubble, and by protecting the fragile—the black majority, the white dissenters, the vision of a rainbow nation—until the old system collapsed under its own contradictions.

None of these were perfect. All had their own shadow sides. But they all demonstrate the same pattern: peace is not achieved by balancing swords. It is achieved by nurturing the social organism back to health.

Part IV – Building the Compass

13. A Compass, Not a Map

Up to this point, I've been describing a diagnostic—a way to see the health of a social body. But a person standing in the middle of a complex situation needs more than a description; they need a tool they can use in real time to orient their own decisions.

That tool is the Cube.

The Cube is simply the three-axis space we already explored: Feedback, Curiosity, Respect. But now imagine it as a physical or mental device—a compass you can consult.

When facing any decision, personal or political, ask:
- Will this action shorten or lengthen the feedback loop between me and those affected?
- Will it open or close the space of curiosity?
- Will it protect or endanger the fragile?

You don't need perfect answers. The act of asking itself begins to rewire your instincts. Over time, you start to feel the pull of the Constructive Attractor and the repulsion of the Destructive Attractor.

The Cube becomes an inner gyroscope.

It doesn't tell you what to believe. It doesn't prescribe a list of rules. It is not an ideology. It's a meta-tool that works with any religion, any culture, any personal philosophy. A Christian, a Muslim, a secular humanist can all use the Cube because it doesn't replace their core commitments; it only helps them navigate the *consequences* of those commitments.

And when used collectively—by a community, an institution, a diplomatic process—the Cube shifts the conversation from "Who is righteous?" to "What are we building together?"

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14. The Art of Self-Regulation

A compass is useless without the practice of navigation.

Using the Cube regularly—daily, weekly, in moments of crisis—is a skill. It's like learning to read the stars. At first it's awkward and abstract. Later, it becomes intuitive.

The practice can take many forms:
- A personal journal where you review the day's decisions against the three axes.
- A family ritual where you discuss a recent conflict and ask together how to move toward more feedback, more curiosity, more respect.
- An institutional meeting that begins not with the agenda, but with a check-in on the health of the group's feedback loops.
- A national dialogue where proposed laws are publicly evaluated on the Cube before a vote.

The point is regularity. You build the muscle of self-regulation by using it. And as with any muscle, it weakens with disuse.

The great exemplars—Jesus, Mohamed, Buddha, Gandhi, King—were not people who achieved a static perfection. They were people who, through relentless practice, aligned themselves so deeply with the Constructive Attractor that their very presence tilted the cone around them. They became living compasses.

We don't have to be exemplars. We just have to practice.

Conclusion – The Necessity of Dice

Einstein famously said, "God does not play dice."
He was rejecting the randomness of quantum mechanics, clinging to a universe that was fully determined, fully knowable, fully safe.

This order is not encoded in the program; it is a physical event, paid for in processor energy. Each such event is a *resolution* of the mutual constraint, and that resolution becomes part of the evolving state, amplified and redistributed by the mixing function.

The critical point: the initial state (all zeros) and the deterministic code do not force any particular sequence of memory-operation interleavings. Numerous sequences are compatible with the initial condition, and each yields a different subsequent state. The static description of the system is therefore underdetermined. The system must run to resolve which sequence actually occurs, and that resolution is a novel, irreproducible act.

4.3 Empirical Results

Two empirical properties confirm that `Seedy64` is a generative topology, not a chaotic one.

Statistical randomness. Output streams from `Seedy64` pass the PractRand test suite up to at least 2^20 bytes and beyond. No algorithmic pattern is detectable by standard statistical batteries.

Exhaustive coverage. When the device is started, allowed to produce only its first 64-bit output, then immediately destroyed and rebuilt, the collected first outputs over thousands of runs uniformly cover the entire 32-bit integer range. The same all-zero start yields, over many fresh runs, every possible small integer. This is not a hidden attractor; it is exhaustive novelty.

These results refute the hypothesis that `Seedy64` is merely amplifying microscopic differences in initial conditions (because the initial condition is digitally identical each time). They demonstrate that the system *creates* its output, paying for it in energy, and that its output cannot be derived by any simulation shorter than the actual run.

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5. The Code and the Machine: Why Generativity Requires a Runtime

The empirical success of `Seedy64` clarifies an essential feature of generative topologies: the self-referential rule set alone is not sufficient. The code, if executed in a perfectly deterministic, sequentially ordered virtual machine that imposes a fixed update order, would produce a repeatable trajectory. Generativity arises from the *coupling* between the self-referential rules and a physical runtime whose behaviour is not fully captured by those rules. In the case of `Seedy64`, the runtime is the CPU, operating system, and memory hierarchy—complex enough to supply irreproducible interleavings, yet not random in the sense of a hardware random number generator. The interleavings are the product of a deterministic but unpredictable physical process (the microarchitecture's internal dynamics), and they are *necessary* for the ring to resolve.

This separation is crucial for what follows. A generative topology is not a piece of software; it is a

2. **A physical runtime environment:** a non-ideal execution medium that forces the rule set to be resolved asynchronously in time, introducing an irreproducible serialization of events that is not determined by the rules alone.

When these two ingredients combine, the system exhibits **active creative novelty**: it generates states that are underdetermined by any static description of the initial conditions and rules. Such states are not merely unpredictable due to hidden variables or measurement error; they are *ontologically new*—they come into being only through the act of resolution.

Generative topologies are not chaotic. A chaotic system can be simulated perfectly if given perfectly precise initial conditions. A generative topology cannot, because the initial condition (the code plus the starting values) does not, even in principle, pick out a unique trajectory. The trajectory is *chosen* by the runtime. Generative topologies are thus a third category, distinct from both deterministic chaos and indeterministic randomness.

The following section provides a minimal, empirically testable instance of such a system.

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4. The Seedy64 Proof: A Minimal Generative Topology

4.1 Construction

`Seedy64` consists of three 64-bit unsigned integer memory cells, `nodes[0]`, `nodes[1]`, `nodes[2]`, initialised to zero. Three operating-system threads are launched, each dedicated to one directed edge of a ring: thread 0 reads `nodes[0]`, transforms the value via a mixing function, and writes the result to `nodes[1]`; thread 1 reads `nodes[1]` and writes to `nodes[2]`; thread 2 reads `nodes[2]` and writes to `nodes[0]`. The mixing function involves bit rotations, XOR operations, and addition with a prime-based accumulator, designed to propagate any bit-flip to all bits within a few iterations. All reads and writes are performed with volatile semantics, forcing actual memory access and preventing compiler reordering. No synchronization primitives (locks, atomics) are used.

The main thread allows the ring to churn for an initialisation period, then samples the XOR of all three nodes at regular intervals. Whenever the sampled value differs from the previous sample, the difference is used to fill a user-provided byte buffer.

4.2 The Generative Mechanism

The ring of dependencies forms a self-referential topology: the next value of `nodes[1]` depends on `nodes[0]`, which depends on `nodes[2]`, which depends on `nodes[1]`. Because the threads run concurrently on a multi-core processor, the order in which reads and writes interleave is determined by the hardware's memory subsystem—cache coherence traffic, bus arbitration, and core scheduling.

But Einstein was wrong, and not just at the quantum level. The universe plays dice in its very structure. Self-reference—the mutual dependence of parts on each other—is built into the fabric of reality. That self-reference means no future is fully determined. Every moment is a genuine resolution of a constraint network, a throw of cosmic dice.

The same is true of human beings. We are a society of societies, a body of bodies, a mind of minds, all mutually dependent, all influencing each other's resolutions. We cannot predict the future. We can only tilt the dice.

Making peace without killing each other is not about enforcing a stalemate. It's about learning to tilt the dice toward the Constructive Attractor. It's about shortening feedback loops, restoring curiosity, and shielding the fragile—every day, in every relationship, in every institution.

The demons of determinism whisper that it's impossible: the hatreds are too old, the interests too entrenched, the cycles too vicious.

The Seedy device whispers back: *From the same zero state, any number can appear.*

The future is not a prison. It's a garden that we tend with our attention.

Let us tend it toward life.

cannot be statically determined.

That same creativity echoes in every living thing and every conscious moment. An embryo, a thought, a decision—each, as we will see, is a generative topology that must run to resolve. The future is not yet written. It never was. And this paper is about why that must be so.

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2. The Demon's Blind Spot: Why Static Descriptions Fail

Laplace's confidence rested on an unexamined assumption: that a complete state description, plus complete laws, forms a well-posed initial value problem. In physical modeling, this works for systems whose dependency graph is a directed acyclic network—each variable's future value depends only on variables at earlier times, never on peers or on itself. But not all systems are so linear. When dependencies form closed loops, the assumption collapses.

Consider three variables, A, B, and C, whose next states are determined simultaneously by each other: $A \leftarrow f(B)$, $B \leftarrow g(C)$, $C \leftarrow h(A)$. This is an algebraic loop. Mathematically, one may search for a fixed point that satisfies all three equations at once; but such a fixed point may not exist, may not be unique, and even if it exists, the process of *finding* it is not captured by the equations themselves. In a physical implementation, there is no instantaneous global solver. The system must evolve, components updating asynchronously, and the order of those updates becomes a determining factor—yet that order is not deducible from the static equations. It must be *provided by time*.

This is the logical gap. The demon, given only the equations and the current values, would have to simulate the system's entire serialization history to know the outcome. But that simulation is itself a physical process that consumes time and energy. The demon cannot leap ahead of the world it is trying to predict without *being* that world. The snapshot is insufficient not because it lacks precision, but because it lacks a dimension: the order of resolution.

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3. Generative Topologies: A Definition

I propose the term **generative topology** to name the class of systems that exploit this gap to produce novelty. A generative topology is defined by two components:

1. **A self-referential rule set:** a set of deterministic update rules whose dependencies contain at least one closed loop, meaning that the next state of some component depends, directly or indirectly, on its own current state via a cycle of mutual reference.

1. Introduction – The Surprise That Built the Universe

Imagine the smartest demon that could ever exist. Laplace's demon knows every particle in the universe, every force, every law. Give it a single perfect snapshot of Now, and it could tell you the entire future and past. For centuries this thought experiment felt unshakeable. If the world is made of deterministic pieces, then everything that ever happens is already folded inside the present, like a movie that has already been shot.

But what if some things simply *aren't there yet*—not because the demon is ignorant, but because the snapshot itself is incomplete? Not missing a detail, but missing something that only *happening* can provide. This is not a gap in our physics. It is a gap in logic, and a tiny computer program has made it impossible to ignore.

Picture three friends playing a strange game. Each must call out a number, but the rule is perverse: Ana's next number depends on what Ben is about to say, Ben's depends on Clara's, and Clara's depends on Ana's. They all must answer *at the same time*. No one can go first. There is no correct answer in advance; the rules don't pick a unique number. Yet if they actually start shouting, a number will emerge, shaped by tiny real□time accidents—a hesitation, a breath, a glance. That number didn't exist before they played. It was *created* by the playing.

This is the heart of a device called `Seedy64`, a ring of three simple threads on a computer chip, designed to probe the nature of creative novelty. The threads depend on one another in exactly such a loop. The machine starts with every value set to zero. No outside randomness is fed in. Yet when it runs, the first output is completely unpredictable; run it again from the same zeros, and you'll eventually get every possible number. It passes the toughest randomness tests. The code is totally deterministic, yet the system behaves as if it is throwing dice. It isn't chaos in the usual sense—chaos amplifies tiny hidden differences, but here the starting point is perfectly identical each time. The system genuinely *generates* new information, paying for it in processor heat. The novelty is not hidden in the initial state; it is born in the act of resolving a mutual stand□off.

Philosophers have long flirted with the idea that time itself creates possibilities. Henri Bergson insisted that the future is not just a rearranged past, but an ongoing invention. The ring gives this a concrete skeleton: some patterns of interdependence have no unique solution on paper. They must be *lived* to be settled. That act of settling is a creative event, a tiny big bang of newness.

Now scale this up. At the beginning of our cosmos, the newborn universe was not a neat line of dominoes waiting to fall. It was a dense, over□excited tangle where everything was coupled to everything else—a ferociously self□referential knot. The Big Bang itself may have been the grandest resolution of a mutual□dependency network. The laws were there, but the outcome was not yet fixed. The first structure—the clumping of matter, the breaking of symmetries—was the universe's first "number," shouted into existence by the sheer necessity of becoming. Dice were not added later by quantum mechanics; they were woven into the beginning because a self□implicating origin

Open Invitation

From Sylvain Saucier
Oppenheimer Park, DTES, Vancouver, Canada

To anyone dreaming of World Peace — those who tend the fragile, who teach the curious, who hold the mirror to our collective conscience — I am writing to invite you to a conversation that I believe our century cannot afford to postpone.

My name is Sylvain Saucier. I am not a priest, a prophet, or a politician. I am a former truck driver, a sometime graphic designer, a student of cybernetics and process philosophy, and a modest resident of Vancouver's Downtown Eastside — a neighbourhood that many call Canada's poorest postal code, but which I call home and the laboratory of my philosophy.

I have spent years developing a simple compass for ethical navigation. I call it **the Cube**. It is not an ideology, not a religion, not a replacement for your traditions. It is a meta-tool — three axes, one attractor — designed to help anyone, from any faith, evaluate the systemic consequences of their choices. The axes are short feedback, radical curiosity, and the protection of the fragile. The direction is what I call the constructive attractor, the (+++) corner where diversity grows and peace is not a truce but a garden.

I have written papers — *Time as Dice*, *The Cube*, *Pax Perspicua*, *Living Governance*, *Energy as Dice* — that try to build a philosophical foundation for a 5000-year peace. But papers are not enough. The real work requires spiritual authority, moral courage, and voices that can reach the hearts of billions. That is you.

What I am proposing is a **gathering of every major tradition** — Christian, Muslim, Jewish, Buddhist, Hindu, Indigenous, and all who hold a sacred responsibility for the human soul — to sit together and discuss not who is most righteous, but what our theologies *produce* when they are scaled into laws, economies, and borders. To ask: *Does this teaching shorten feedback loops or lengthen them? Does it open curiosity or close it? Does it protect the fragile or sacrifice them for the sake of order?*

I had dreamed of hosting this conversation in Gaza. It would have been fitting — a land sacred to three faiths, a wound in the body of humanity, a place where the need for a new kind of peace is most urgent and most visible. But the physical reality of war and insecurity makes that impossible for now. Perhaps one day.

For now, I ask you to come to me.

I live in the **Downtown Eastside of Vancouver, British Columbia**. It is a place of extreme fragility — of addiction, poverty, mental illness, and also of extraordinary resilience, mutual care, and daily acts of grace. It is a microcosm of the destructive attractor and the constructive attractor, side by side, on every block. If you want to talk about systemic morality, this soil is honest. It does not hide its wounds, and it does not pretend to be righteous.

I have no palace to receive you. I have no budget, no institution, no security detail. What I have is a philosophy, a compass, and a conviction that the spiritual leaders of the world are capable of more than competing for followers. You are capable of building a shared immune system for the human spirit — a way to keep the garden tended even as the energy of empire fades.

I ask you to come to Vancouver. Sit with me in the cold rain. Walk these alleys. Meet the people who survive here. Then let us sit in a circle — perhaps in a community hall, perhaps in a church basement — and begin the conversation that could define Phase 1 of the Great Correction.

I do not expect all of you to agree. I do not expect anyone to abandon their revelation. I expect only that you bring your scriptures, your doubts, your compassion, and your willingness to be questioned by a compass that asks nothing of your faith except that it be *kind in its outcomes*.

If you are willing, I will make myself available. I will prepare the space as best I can. I will invite the local community to witness, and I will ensure that the conversation is recorded and shared freely with the world — because this is not a closed summit. It is a public act of hope.

The Cube is a common inheritance. I release it to you. Take it, test it, break it if you must. But if it holds, use it. Let us shift the global dialogue from a contest of righteousness to an inquiry into consequences. Let us prove that the world's spiritual traditions, working together, can face the coming energy descent, the rise of authoritarian attractors, and the suffering of the fragile — not with swords, but with a shared compass and a shared resolve.

The Downtown Eastside awaits. I await.

With humility and stubborn hope,

Sylvain Saucier
Philosopher
Vancouver, British Columbia, Canada

Meet me at Oppenheimer park, ask around for Sylvain "Joe" the philosopher.

Generative Topologies: Self-Referential Networks and the Origin of Novelty

Author: Sylvain Saucier
Repository: github.com/sysaulab/softrng

Abstract

For centuries, Laplace's demon has embodied the belief that a perfect snapshot of the present, combined with deterministic laws, yields the entire future. This paper presents a logical, not probabilistic, refutation of that belief. I introduce the concept of a *generative topology*: a system whose internal dependencies form closed loops, such that its evolution is underdetermined by any static description. When such a topology is placed in a physical runtime—a computer, a cell, a brain—the system must actively resolve its own mutual constraints in time, generating states that did not exist, even implicitly, in its initial specification. I provide an empirical minimum viable proof: the `Seedy64` device, a deterministic ring of three threads that, from an all-zero start, produces statistically ideal randomness and exhaustively covers the 32-bit integer space. The device demonstrates that self-referential code, coupled to a real execution environment, forms a generative topology whose output cannot be predicted by any simulation shorter than the run itself. This principle is then extended: the fertilised egg is a generative topology of genome and cytoplasm; the conscious brain is a generative topology of neural loops and sensory input; the Big Bang itself was the ultimate generative topology. The universe, life, and mind do not unfold pre-written scripts—they actively create novelty. Generative topologies are the grammar of becoming.
